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LLM-Powered Query Expansion for Enhancing Boundary Prediction in Language-Driven Action Localization

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Abstract

Language-driven action localization in videos requires not only semantic alignment between language query and video segment, but also prediction of action boundaries. However, the language query primarily describes the main content of an action and usually lacks specific details of action start and end boundaries, which increases the subjectivity of manual boundary annotation and leads to boundary uncertainty in training data. In this paper, on one hand, we propose to expand the original query by generating textual descriptions of the action start and end boundaries through LLMs, which can provide more detailed boundary cues for localization and thus reduce the impact of boundary uncertainty. On the other hand, to enhance the tolerance to boundary uncertainty during training, we propose to model probability scores of action boundaries by calculating the semantic similarities between frames and the expanded query as well as the temporal distances between frames and the annotated boundary frames. They can provide more consistent boundary supervision, thus improving the stability of training. Our method is model-agnostic and can be seamlessly and easily integrated into any existing models of language-driven action localization in an off-the-shelf manner. Experimental results on several datasets demonstrate the effectiveness of our method.

Keywords LLM-powered query expansion · Enhancing boundary prediction · Language-driven action localization

1 Introduction

Language-driven action localization, also known as video moment retrieval (Gao & Xu, 2021; Wu et al., 2021; Zeng et al., 2021; Tang et al., 2021) or temporal sentence grounding (Yuan et al., 2019; Liu et al., 2021b, c), aims to localize the temporal action segment semantically aligned with a natural language query in an untrimmed video. It has received increasing attention in recent years and has wide applications in many downstream tasks such as video question answering (Sun et al., 2021) and video editing (Zhang et al., 2022b). Recent studies (Gao et al., 2017; Liu & Zhou, 2023; Liu et al., 2021a; Zhang et al., 2020b) have achieved remarkable progress in language-driven action localization by aligning the video segment with the language query (Ghosh et al., 2019; Zhang et al., 2023; Liu et al., 2022a, 2024) and predicting the desired action boundaries using rigid boundary annotations as supervision (Zhang et al., 2020a; Liu & Hu, 2022; Liu et al., 2022b, 2021b).

However, the language query primarily describes the main content of an action, and lacks details of start and end boundaries. Different human annotators may have different subjective interpretations of action boundaries, and give inconsistent boundary annotations for the same type of action in different videos, resulting in boundary uncertainty. As shown in Fig. 1, for the language query of "person eats sandwich" that describes the same action "eat", the start boundary annotation of one video (a) refers to the motion "place hand on food plate", which that of another video (b) refers to the motion "put food into the mouth". Therefore, learning to localize accurate boundary timestamps from inconsistent boundary annotations is highly challenging and is still under exploration.

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Fig. 1 Illustration of boundary uncertainty for the same language query. The orange text represents the motion of the start boundary annotation and the blue text represents the motion of the end boundary annotation

To address this challenge, we propose LLM-powered query expansion, which leverages a large language model (LLM) to generate detailed textual descriptions of action start and end boundaries to expand the original query. Specifically, we first design several prompts, such as *"Please describe the beginning and ending process in one sentence of the following action [ACTION], The description you generate cannot contain any objects that are not present in the action"*, for the LLM to generate fine-grained descriptions about motions, poses, or temporal differences that distinguish boundaries. Then, we propose a query-guided temporal modeling module that incorporates the expanded query into visual representation learning. By injecting explicit boundary cues, the model is able to extract discriminative features that capture boundary details the original query might overlook, thereby refining the "input" perception of action boundaries and reducing the impact of uncertainty.

On the other hand, to further enhance the tolerance to boundary uncertainty during training, we propose a boundary probability modeling module that estimates the probability of each video frame being an action boundary based on the expanded query. This provide the "output" supervision with probabilistic flexibility, enabling the model to map discriminative features to action boundaries. Specifically, for each frame, we model the probability of being an action boundary by calculating its semantic similarity to the expanded query and its temporal distance to the labeled boundary frame. The semantic similarity ensures that frames relevant to the action described by the expanded query are prioritized, while the temporal distance reflects the intuition that frames closer to the annotated boundary are more likely to correspond to the action boundary. The probability scores, instead of rigid annotations, reduce over-reliance on a single annotated boundary by providing more flexible and reliable supervision, thereby better capturing motion-related information near the boundary. In addition, the clear guidance of the expanded query ensures that the probability scores are more consistent across similar actions, further providing more consistent supervision and improving the stability of training.

The main contributions can be summarized as follows:

- We propose a query expansion method that expands the language query through LLM to capture more details of action boundaries, effectively reducing the impact of boundary uncertainty in action localization.
- We propose a boundary probability modeling module that utilizes the expanded query to transform rigid boundary annotations into probability scores for training, successfully improving the robustness to boundary uncertainty.
- Our method can be seamlessly and easily integrated into existing models in an off-the-shelf manner. Extensive experiments on five state-of-the-art models across three benchmark datasets demonstrate the effectiveness of our method.

2 Related Work

Existing methods of language-driven action localization can be broadly categorized into two groups: proposal-based methods and proposal-free methods. The proposal-based methods first generate segment proposals using sliding windows, proposal generation, or anchor-based methods and then rank them according to the query. Through exhausting proposal generation and directly aligning the semantics of the query and proposals, the proposal-based methods achieve better performance. Early methods such as CTRL (Gao et al., 2017) and MCN (Hendricks et al., 2017) use sliding windows of varying scales to generate candidate video segments, while later methods such as SPN (Xu et al., 2017), QSPN (Xu et al., 2019), and SAP (Chen & Jiang, 2019) adopt segmented proposal networks tailored to specific queries. Recent studies, including 2D-TAN (Zhang et al., 2020b), RaNet (Gao et al., 2021), and CDN (Wang et al., 2022), utilize two-dimensional feature maps to model relationships across segments of different durations. MIPGN (Fang et al., 2025) proposes a multi-modal integrated proposal generation network to guide adaptive proposal generation, incorporates object and motion tags for richer multi-modal representations, and employs dynamic negative proposal mining to improve retrieval accuracy. VGCI (Lv & Su, 2025) introduces a variational inference framework for weakly supervised video moment retrieval, modeling a global clue as a latent Gaussian variable to guide query reconstruction and enhance video-query alignment.

The proposal-free methods first learn cross-modal interactions between query and video, and then predict the frame-wise boundary probabilities (Zhang et al., 2020a; Liu

& Hu, 2022; Liu et al., 2022b, 2021b) or regress the start and end boundary frames of the target segment (Ghosh et al., 2019; Zhang et al., 2023; Liu et al., 2022a, 2024). Zhang et al. (2022a) search for the target action within a highlighted region in a span-based question-answering framework. Mun et al. (2020) extract the implicit semantic information from local to global by a sequential query attention module. Sun et al. (2022) leverage fine-grained intra-modality clues to explore deeper inter-modality information. UniVTG-NA (Flanagan et al., 2025) introduces a negative-aware video moment retrieval framework that distinguishes between in-domain and out-of-domain negative queries, enabling robust rejection of irrelevant queries. SSTr (Huo et al., 2025) proposes a Skim-and-Scan Transformer architecture, leveraging dual-granularity feature interaction and dual-supervision strategies to model cross-video semantics at multiple temporal scales. Recently, DETR-based methods (Xiao et al., 2024; Yang et al., 2024; Jang et al., 2023; Moon et al., 2023a) have been proposed to apply the popular DETR (Carion et al., 2020) architecture into language-driven action localization, which have achieved remarkable performance. BM-DETR (Jung et al., 2025) proposes a background-aware moment detection transformer that leverages contrastive learning with negative queries to better utilize surrounding context and alleviate weak alignment issues in video moment retrieval. W2W (Liu et al., 2025) proposes a progressive "what and where" framework, combining cross-modal semantic alignment via Initial Semantic Projection and Clip Semantic Mining with moment-level temporal context modeling using a dual-branch Multi-Context Perception module.

The most related work to our method are Otani et al. (2020), Pan et al. (2022) and Huang et al. (2022), which focuses on solving the problem of boundary annotation uncertainty. Otani et al. (2020) quantitatively study the uncertainty in the temporal annotations by collecting multiple boundaries for the same action from different annotators, but do not explicitly propose a solution. Pan et al. (2022) adopt Gaussian smoothing to generate soft labels for each boundary, and Huang et al. (2022) propose elastic moment bounding between annotated boundaries and predicted boundaries to generate pseudo labels. These methods heuristically expand the rigid start and end boundaries into soft labels, considering only temporal visual differences. In this paper, we leverage a powerful LLM to exploit more explicit interpretations for each boundary, thereby alleviating the ambiguity of boundary definitions and reducing the uncertainty of boundary annotations.

3 Our Method

In this section, we present the proposed framework for language-driven action localization under boundary uncertainty. We first provide a high-level Overview of the pipeline in Sect. 3.1, Sect. 3.2 defines the Base Model we use. To enrich the semantic boundary cues, the LLM-powered Query Expansion process is introduced in Sect. 3.3. The proposed Query-guided Temporal Modeling and Boundary Probability Modeling are introduced in Sects. 3.4 and 3.5, respectively.

3.1 Overview

Given an untrimmed video V and a language query Q, we present the video as $V = \{v_t\}_{t=1}^{T}$, where v_t is the t-th frame, and T is the number of total frames. The language query with N words is denoted as $Q = \{q_n\}_{n=1}^{N}$. Our task aims to locate the start and end timestamps (τ_s, τ_e) of the action described by the language query Q in the untrimmed video V.

As shown in Fig. 2, we first prompt the LLM to generate fine-grained textual descriptions of start and end boundaries as expanded queries, respectively. We then extract the video features $F_v \in \mathbb{R}^{T\times D}$ for V, and extract the start query features $F_s \in \mathbb{R}^{N_s\times D}$, the original query features $F_q \in \mathbb{R}^{N\times D}$, the end query features $F_e \in \mathbb{R}^{N_e\times D}$ for Q_s, Q, and Q_e, respectively, using the visual and text encoders of the base model $\mathcal{M}$. We propose a query-guided temporal modeling module to get enhanced query features $F_g \in \mathbb{R}^{N\times D}$ and visual features $F_v' \in \mathbb{R}^{N\times D}$, which are then fed into the localization head of $\mathcal{M}$ to obtain predictions. We also design a boundary probability modeling module to generate probability scores of action boundaries by combining cross-modal semantic similarity and inter-frame temporal distance to refine the predictions.

3.2 Base Model

Our proposed method is designed to be model-agnostic. To demonstrate its effectiveness, we build upon a kind of representative DETR-based framework(e.g., Lei et al. 2021; Moon et al. 2023b), as our base model. For structural clarity, this framework is abstracted into two core components consistent with our overall pipeline: Visual and Text Encoders for input representation, and a Localization Head for task-specific outputs. In our full pipeline, the Query-guided Temporal Modeling module (Sec. 3.4) is integrated as an intermediate enhancement between these two components to refine the feature representations.

Visual and Text Encoder. Following the problem formulation, we first obtain raw video and query features using pretrained backbones. Specifically, F_v^{raw} is formed by concatenating SlowFast (Feichtenhofer et al., 2019) and CLIP (Radford et al., 2021) visual features, while $\{F_q, F_s, F_e\}^{raw}$ are extracted via the CLIP text encoder. To bridge the dimensionality gap between these heterogeneous backbones and the subsequent Transformer, we then employ

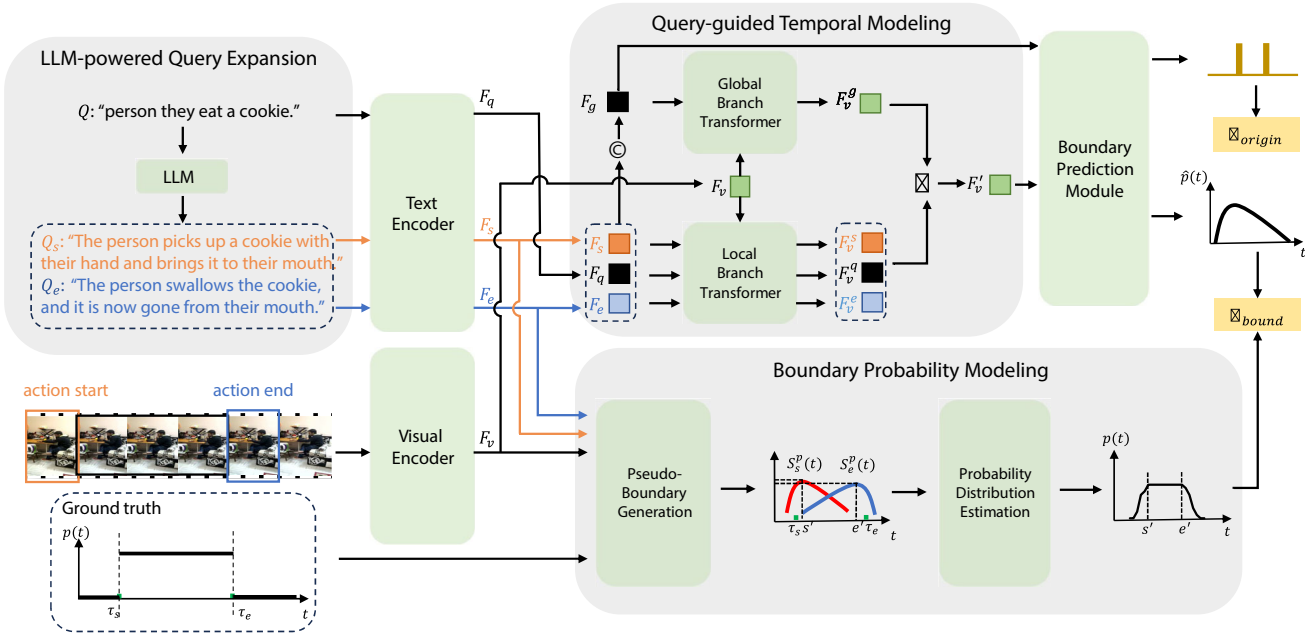


Fig. 2 Overview of the proposed action localization framework. The pipeline first leverages a large language model to perform text expansion, generating detailed sentences that describe the beginning and ending stages of the target action based on the original content-focused query. These expanded boundary cues, along with the visual video, are processed by the base model to obtain initial text and video features. To refine the multimodal representations, we introduce a query-guided temporal modeling module that fuses the detailed start and end descriptions with the video features, employing global and local branches to capture holistic context and stage-specific boundary details. Finally, a boundary probability modeling module is proposed to estimate the likelihood of action boundaries across the temporal sequence, converting rigid ground-truth intervals into soft probability scores for more robust supervision during training

2-layer MLPs as projection layers to map all raw features into a unified hidden dimension D:

$$F_i = \text{MLP}_i(F_i^{raw}), \quad i \in \{v, q, s, e\}. \tag{1}$$

Localization Head. This module adopts a Transformer-based encoder-decoder architecture to process the interacted features and predict temporal boundaries. In the base model, the projected video features F_v and query features F_q are integrated by simply concatenating them along the sequence length dimension to form a joint input sequence:

$$E_{input} = [F_v; F_q] \in \mathbb{R}^{(T+N)\times D}. \tag{2}$$

The input features E_{input} are then fed into a stack of T Transformer encoder layers, which utilize multi-head self-attention to generate contextualized representations $E_{enc} \in \mathbb{R}^{(T+N)\times D}$. The decoder then refines M learnable moment queries through cross-attention over E_{enc}, producing moment-aware embeddings $E_{dec} \in \mathbb{R}^{M\times D}$.

To generate final predictions, E_{enc} is used to compute clip-level saliency scores $S \in \mathbb{R}^T$ via a linear layer. Simultaneously, E_{dec} is passed through task-specific heads: a 3-layer FFN to regress normalized temporal boundaries $\hat{m} = (\hat{c}, \hat{w})$ and a linear classifier to produce foreground/background scores.

Training Loss. To ensure a fair comparison and maintain model-agnosticism, our framework inherits the original optimization objectives of the chosen base model $\mathcal{M}$, as described in their respective works (Lei et al., 2021; Moon et al., 2023b). Specifically, the joint origin loss $\mathcal{L}_{origin}$ is formulated as a combination of three functional components:

$$\mathcal{L}_{\text{origin}} = \mathcal{L}_{\text{loc}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{sal}}, \tag{3}$$

where $\mathcal{L}_{loc}$ regresses the temporal boundaries, $\mathcal{L}_{cls}$ performs foreground-background classification, and $\mathcal{L}_{sal}$ facilitates saliency ranking.

3.3 LLM-Powered Query Expansion

In language-driven action localization, the input language query primarily describes the main content of an action and usually neglects details of start and end boundaries, which increases subjective interpretations by human annotators, leading to inconsistent boundary annotations. To reduce the impact of boundary uncertainty, we leverage LLMs to expand the original query by explicitly describing more detailed information of action boundaries, such as motions, human

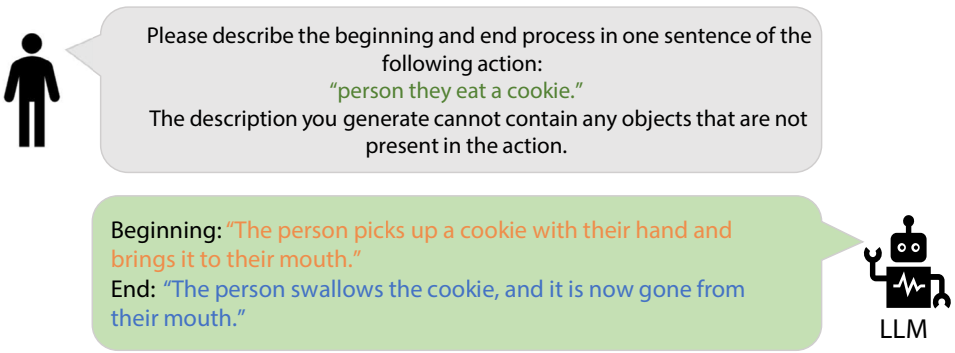


Fig. 3 An example of the designed question and the corresponding answers from LLM

poses, and temporal differences that distinguish boundaries, providing more fine-grained boundary cues for localization.

We design a prompt for the LLM (LLaMa3-8B (Dubey et al., 2024) is used in this paper) to generate two expanded queries: one query describing the action start and the other describing the action end. The prompt is crafted to guide the LLM in producing descriptions that are semantically aligned with the original query and explicitly focus on the temporal boundaries of action, such as *"Please describe the beginning and ending process in one sentence of the following action [ACTION]"*. To reduce the potential hallucination problem of LLMs and maintain the consistency of the expanded queries for similar actions, we instruct the LLM: *"The description you generate cannot contain any objects that are not presented in the action"*. Furthermore, to ensure conciseness and minimize textual noise, we require the LLM to describe each process in exactly one short sentence and strictly limit each sentence to no more than 20 words. This ensures that the expanded queries align with the semantic content described by the original query while providing detailed boundary-specific information. Figure 3 provides an example of the designed prompt and the corresponding outputs.

By providing finer boundary details and enriching the semantic context for localization, the expanded queries help capture fine-grained temporal information that is overlooked by the original coarse query. For the expanded start query Q_s and end query Q_e, we use the text encoder of the base model to extract their corresponding feature F_s and F_e, respectively, where D is the feature dimension.

3.4 Query-Guided Temporal Modeling

To effectively leverage the semantic guidance from expanded queries, we propose the Query-guided Temporal Modeling module. As illustrated in the overall pipeline, the module acts as a plug-and-play enhancement layer inserted before the Localization Head. By injecting explicit boundary cues into the visual representation learning, this module is able to extract discriminative features that capture boundary details the original query might overlook, thereby refining the "input" perception of action boundaries. Specifically, the module employs a dual-branch architecture: local branch that focuses on fine-grained boundary representations at specific action stages, and a global branch that captures the overall temporal structure of the entire sequence.

Local Branch. The local branch enhances the video feature $F_v \in \mathbb{R}^{T\times D}$ by performing fine-grained temporal modeling for each action stage, using the start query feature $F_s \in \mathbb{R}^{N_s\times D}$, the original query feature $F_q \in \mathbb{R}^{N_q\times D}$, and the end query feature $F_e \in \mathbb{R}^{N_e\times D}$ as guidance, respectively. Specifically, F_v is treated as the *query*, while F_s, F_q, or F_e serves as the *key* and *value* in the Multi-Head Cross-Attention (MHCA) mechanism, resulting in locally enhanced features $F_v^s, F_v^q, F_v^e \in \mathbb{R}^{T\times D}$:

$$F_v^i = \mathrm{MHCA}(F_v, F_i, F_i), \quad i \in s, q, e, \tag{4}$$

which encode temporal details specific to the start, main body, and end stages of the action.

Global Branch. The Global Branch enhances F_v by capturing holistic temporal dependencies across the entire sequence, using the concatenated query feature $F_g = [F_s; F_q; F_e] \in \mathbb{R}^{(N_s+N_q+N_e)\times D}$ as guidance. Specifically, F_v serves as the *query*, and F_g is used as *key* and *value* to obtain the globally enhanced feature $F_v^g \in \mathbb{R}^{T\times D}$:

$$F_v^g = \mathrm{MHCA}(F_v, F_g, F_g). \tag{5}$$

Finally, the globally enhanced video feature and the locally enhanced video features are weighted fused to obtain the final video feature F_v':

$$F_v' = aF_v^g + b(F_v^s + F_v^q + F_v^e), \tag{6}$$

where a and b are hyper-parameters.

Through this dual-branch refinement, the initial features F_v and F_q are transformed into boundary-discriminative visual features F_v' and enhanced query features F_g. While the base model directly utilizes the raw concatenated features

$E_{input} = [F_v; F_q]$ (Eq. 2), the integrated framework replaces it with the enhanced joint sequence $E'_{input} = [F'_v; F_g]$ as the final input for the subsequent Localization Head.

3.5 Boundary Probability Modeling

Although the expanded queries effectively reduce boundary uncertainty by providing detailed boundary descriptions, they cannot completely eliminate the uncertainty caused by human annotators' subjective interpretations of action boundaries. To this end, we propose a boundary probability modeling module that leverages the rich semantic information from expanded queries to estimate probability scores of potential action boundaries. By using these scores as flexible "output" supervision instead of rigid annotations, the model is able to focus on a boundary region rather than a single frame to map the discriminative features to action boundaries. This provides the probabilistic flexibility to effectively enhance the tolerance to boundary uncertainty during training. In addition, the probability scores of boundaries are more consistent between similar actions, further providing more consistent supervision and improving the stability of training.

The boundary probability modeling module consists of two steps: a pseudo boundary generation step that identifies the most confident action boundaries relevant to the expanded queries, and a probability score estimation step that assigns likelihoods to frames based on their relevances to the pseudo boundaries and the expanded queries, converting rigid annotations into probabilities as supervision for training.

Pseudo Boundary Generation. Since the action description in the expanded query may occur before or after the annotated boundaries, we introduce pseudo boundaries around the ground-truth boundaries, based on two key factors: temporal distance and semantic similarity, to identify the most confident boundaries. The temporal distance emphasizes frames close to the ground-truth boundaries, and the semantic similarity prioritizes frames that align with the boundaries described by the expanded query.

Specifically, for each frame i, we calculate its scores $S_s^p(i)$ and $S_e^p(i)$ by

$$\begin{aligned} S_s^p(i) &= \mathrm{sim}(F_{v,i}, F_s) - \alpha \cdot \mathrm{dis}(i, \tau_s), \\ S_e^p(i) &= \mathrm{sim}(F_{v,i}, F_e) - \alpha \cdot \mathrm{dis}(i, \tau_e), \end{aligned} \tag{7}$$

where $F_{v,i}$ represents the visual feature of the i-th frame, F_s and F_e represent the start query feature and the end query feature, respectively, $\mathrm{sim}(,\cdot,)$ represents the semantic similarity calculated by cosine similarity, $\mathrm{dis}(,\cdot,)$ represents the temporal distance, normalized by the total video length T, i.e., $\mathrm{dis}(i, \tau_s) = |i - \tau_s|/T$, and α is a hyper-parameter to balance two terms. Then, the pseudo boundaries are determined by maximizing the scores:

$$\begin{aligned} \tau'_s &= \arg\max_i S_s^p(i), \\ \tau'_e &= \arg\max_i S_e^p(i). \end{aligned} \tag{8}$$

Probability Score Estimation. Using the pseudo boundaries (τ'_s, τ'_e), we convert rigid boundary annotations into probability scores to better capture boundary uncertainty. Frames that are closer to the pseudo boundaries and show higher semantic similarities to the expanded queries are assigned with higher probabilities of being action boundaries.

Specifically, for each frame $i < \tau'_s$, we calculate its score $S_s(i)$ based on its temporal distance to the pseudo boundary τ'_s and its semantic similarity to the start query Q_s:

$$S_s(i) = \mathrm{sim}(F_{v,i}, F_s) - \alpha \cdot \mathrm{dis}(i, \tau'_s). \tag{9}$$

Based on these scores, we generate a probability score that focuses on the frames that are most likely to be action boundaries. To achieve this, we first filter out frames with scores below a predefined threshold τ, keeping only those frames with high relevance to the action boundary:

$$S'_s(i) = \begin{cases} 0, & S_s(i) < \tau, \\ S_s(i), & S_s(i) \geq \tau. \end{cases} \tag{10}$$

Then, we generate a probability score of the i-th frame being the start boundary by applying min-max normalization to $S'_s(i)$:

$$p_s(i) = \begin{cases} \dfrac{S'_s(i) - S'_{s,\min}}{S'_{s,\max} - S'_{s,\min}}, & S'_s(i) > 0, \\ 0, & S'_s(i) = 0, \end{cases} \tag{11}$$

where $S'_{s,\max} = \max\{S'_s(i) \mid S'_s(i) > 0\}$ and $S'_{s,\min} = \min\{S'_s(i) \mid S'_s(i) > 0\}$. The same process is applied to generate a probability score $p_e(i)$ of the i-th frame being the end boundary, where $i > \tau'_e$.

Finally, the rigid boundary annotations are converted into probability scores by

$$\boldsymbol{p}(i) = \begin{cases} p_s(i), & 0 \leq i < \tau'_s, \\ 1, & \tau'_s \leq i \leq \tau'_e, \\ p_e(i), & \tau'_e < i \leq T. \end{cases} \tag{12}$$

Loss Function. We feed the enhanced video feature F'_v and the query feature F_g into the localization head of the base model $\mathcal{M}$, denoted as $\mathcal{M}_{lh}$, to predict the probability $\hat{\boldsymbol{p}}(i)$ of a frame i being in the action segment:

$$\hat{\boldsymbol{p}}(i) = \mathcal{M}_{lh}(F'_v, F_g). \tag{13}$$

During training, we introduce a boundary loss to minimize the discrepancy between the predicted probability score $\hat{\boldsymbol{p}}(i)$ and the generated probability score $\boldsymbol{p}(i)$ in Eq. (12), using the cross-entropy loss over all frames:

$$\mathcal{L}_{\text{bound}} = \Sigma_{i=1}^{T} \text{Cross-Entropy}(\boldsymbol{p}(i), \hat{\boldsymbol{p}}(i)). \tag{14}$$

The final loss function is computed as

$$\mathcal{L} = \mathcal{L}_{\text{bound}} + \mathcal{L}_{\text{origin}}, \tag{15}$$

where $\mathcal{L}_{\text{origin}}$ is the original loss that the base model uses.

4 Experiment

In this section, we conduct extensive experiments to evaluate the performance of our proposed method. We first introduce the Datasets and Experimental Settings in Sects. 4.1 and 4.2, respectively. A comprehensive Comparison with existing methods is presented in Sects. 4.3 to demonstrate our competitive results. To analyze the contribution of each component, we perform detailed Ablation Studies in Sects. 4.4. Furthermore, a User Study on the quality of generated queries is provided in Sects. 4.5 to assess the effectiveness of our LLM-powered expansion. We also provide Qualitative Results in Sects. 4.6 to visualize the localization performance of our framework. Finally, the efficiency analysis is presented in Sects. 4.7.

4.1 Datasets

Qvhighlights (Lei et al., 2021) is a recently introduced dataset that supports both language-driven action localization and highlight detection. The dataset contains 10,148 videos from YouTube, covering a wide range of content, and includes 10,310 text queries associated with 18,367 annotated moments, with an average of 1.8 moments per query. Different from other datasets that typically have one-to-one query-action pairs, QVHighlights provides a more complex real-world scenario with multiple actions per query. Since the official test set does not have ground-truth annotations, the testing is performed on the official validation set.

Charades-STA (Gao et al., 2017) is constructed by extending the Charades dataset (Sigurdsson et al., 2016) for language-driven action localization. It contains 6670 videos and 16,124 queries, where 12,404 query-action pairs are used for training and 3720 for testing. The average video duration is 30.59 s, and each video contains 2.41 annotated moments, and the moment has an average duration of 8.09 s.

TACoS (Regneri et al., 2013) consists of videos selected from the MPII Cooking Composite Activities video corpus (Rohrbach et al., 2012). It contains 18,818 video-query pairs of different cooking activities. Each video contains 148 queries on average, some of which are annotations of short video clips.

4.2 Experimental Settings

Evaluation Metrics. Following the previous work (Moon et al., 2023b; Nishimura et al., 2024), we adopt Recall1@IoU μ and mean average precision (mAP) as evaluation metrics: (1) R1@μ, which denotes the percentage of top-1 predicted moment having IoU with ground truth larger than threshold μ. (2) mAP, which denotes the mean Average Precision and evaluates the overall performance by averaging the precision values of different IoU thresholds.

Implementation Details. We use LLaMa3-8B (Dubey et al., 2024) as the LLM to generate expanded queries. Both the Local and Global branches utilize a single-layer Transformer with 8 attention heads. The hyper-parameter α is set to 1, while a and b in Eq.(6) are both set to 1. The threshold τ in Eq.(10) is set to 0.8. To ensure a fair comparison, other training configurations (e.g., learning rate, optimizer, and batch size) strictly follow the default settings of the respective base models. We conduct experiments on a single NVIDIA RTX 4090 GPU.

4.3 Comparison with Existing Methods

We conduct experiments by integrating our modules into several existing methods (base models), including QD-DETR (Moon et al., 2023b), Eatr (Jang et al., 2023), TaskWeave (Yang et al., 2024), UVCOM (Xiao et al., 2024), and CG-DETR (Moon et al., 2023a). The comparison results on the Qvhighlights, Charades-STA, and TACoS datasets are shown in Table 1. The results of the original methods are based on our reproduction following Nishimura et al. (2024). From the results, we observe that all the methods integrated with our moduels consistently achieve better performance on all three datasets, highlighting the effectiveness of query expansion and probability score modeling in action boundary prediction.

4.4 Ablation Study

To further evaluate different components of our method, we conduct ablation studies using QD-DETR as base model.

Evaluation of Different Modules. To evaluate the effectiveness of the proposed modules, we conduct ablation studies across different datasets by removing them separately. The experimental results are shown in Table 2, where QTM and BPM denote the query-guided temporal modeling and boundary probability modeling modules, respectively. From the results, we observe that both modules

Table 1 Results of different existing methods integrated with our modules on the Qvhighlights, Charades-STA and TACoS datasets

Method	Qvhighlights			Charades-STA			TACoS		
	R1@0.5	R1@0.7	mAP	R1@0.5	R1@0.7	mAP	R1@0.5	R1@0.7	mAP
QD-DETR (Moon et al., 2023b)	63.16	46.39	40.39	58.17	36.94	34.75	38.17	20.72	20.20
QD-DETR+Ours	**64.13**	**47.10**	**41.16**	**59.71**	**38.58**	**35.97**	**40.83**	**24.04**	**22.49**
Eatr (Jang et al., 2023)	58.26	41.61	36.58	55.67	33.28	33.95	31.69	16.27	16.92
Eatr+Ours	**60.13**	**43.23**	**37.91**	**56.83**	**33.95**	**34.64**	**34.12**	**16.92**	**18.04**
TaskWeave (Yang et al., 2024)	64.26	48.17	44.20	56.85	34.44	35.37	38.52	20.82	19.99
TaskWeave+Ours	**65.31**	**50.00**	**44.90**	**58.44**	**36.51**	**36.84**	**39.20**	**21.5**	**20.70**
UVCOM (Xiao et al., 2024)	64.45	49.81	43.31	57.18	35.43	35.60	40.20	23.30	22.11
UVCOM+Ours	**65.31**	**51.23**	**44.27**	**58.84**	**36.83**	**36.60**	**40.71**	**24.59**	**23.05**
CG-DETR (Moon et al., 2023a)	65.48	50.97	44.03	57.71	35.61	35.07	39.79	24.89	23.28
CG-DETR+Ours	**66.52**	**52.19**	**44.60**	**58.87**	**36.77**	**36.33**	**40.27**	**25.31**	**23.77**

The best results are highlighted in bold

Table 2 Results of different proposed modules with QD-DETR across three datasets. "QTM" and "BPM" denote Query-guided temporal modeling and boundary probability modeling, respectively

Module		Charades-STA			QVHighlights			TACoS		
QTM	BPM	R1@0.5	R1@0.7	mAP	R1@0.5	R1@0.7	mAP	R1@0.5	R1@0.7	mAP
		58.17	36.94	34.75	63.16	46.39	40.39	38.17	20.27	20.91
✓		59.06	37.87	35.15	63.79	46.71	40.91	39.37	22.39	21.45
	✓	59.33	38.25	35.41	63.60	46.82	40.57	40.04	22.14	21.75
✓	✓	**59.71**	**38.58**	**35.97**	**64.13**	**47.10**	**41.16**	**40.83**	**24.04**	**22.49**

The best results are highlighted in bold

Table 3 Results of different LLMs for query expansion with QD-DETR on the Charades-STA dataset

LLM	R1@0.5	R1@0.7	mAP
Noisy query	59.23	38.42	35.91
Qwen	59.51	38.10	36.13
LLaMa2-7B	59.46	38.28	35.81
LLaMa2-13B	59.30	38.31	**36.22**
LLaMa3-8B	**59.71**	**38.58**	35.97

The best results are highlighted in bold

Table 4 Ablation study of different prompt designs on TACoS dataset. "Ours" denotes the default prompt used in the paper

Prompt variant	R1@0.5	R1@0.7	mAP
Prompt-object	39.42	21.92	21.08
Prompt-concise	40.61	23.99	**22.76**
Ours	**40.83**	**24.04**	22.49

The best results are highlighted in bold

independently contribute to improving performance across all metrics. The combination of the two modules achieves the best results, demonstrating their complementary roles in mitigating boundary uncertainty and enhancing localization accuracy.

Evaluation of LLMs for Query Expansion. To evaluate the impact of LLMs for query expansion on the performance, we use Qwen (Bai et al., 2023), LLaMa2-7B (Touvron et al., 2023), LLaMa2-13B (Touvron et al., 2023) for comparison, and the results are shown in Table 3. It is interesting to observe that different LLMs have little impact on performance, and all LLMs have similar R1@0.5, R1@0.7 and mAP scores. Among them, LLaMa3-8B achieves the best overall performance.

In addition, we evaluate the robustness of our method to noisy expanded queries by randomly swapping the start and end queries for a portion of the samples. The results under the "Noisy query" setting show only a slight drop in performance, suggesting that our method remains stable even when the expanded queries contain a certain degree of noise.

Evaluation of Prompt for Query Expansion. To evaluate the impact of prompt engineering on performance, we compare our default prompt with two variants: *Prompt-Object*, which allows the LLM to freely describe objects without constraint, and *Prompt-Concise*, which imposes a stricter length constraint of 10 words per sentence. The results are shown in Table 4. It is observed that the inclusion of additional objects in *Prompt-Object* leads to a performance drop, specifically decreasing R1@0.5 by 1.41% and R1@0.7 by 2.12%. This suggests that hallucinated objects not present in the original action introduce semantic noise

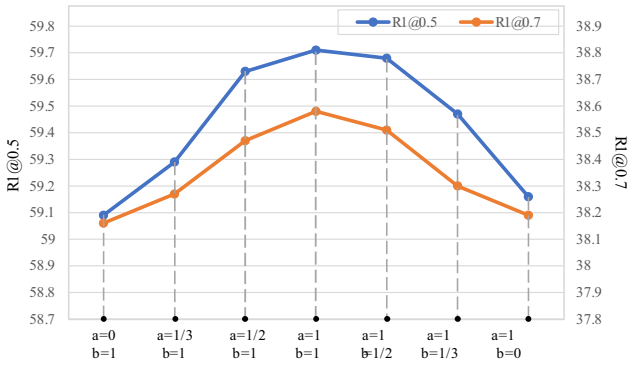


Fig. 4 Results of different values of hyperparameters (a and b) for the global and local branches with QD-DETR on the Charades-STA dataset

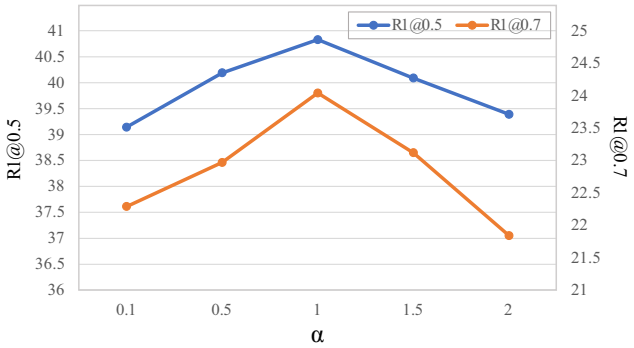


Fig. 5 Results of different values of the hyperparameter (α) for the global and local branches with QD-DETR on the Charades-STA dataset

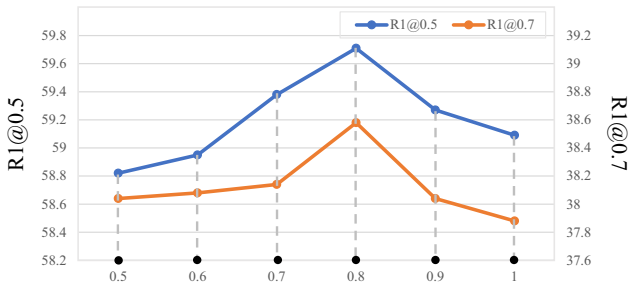


Fig. 6 Results of different values of the hyperparamete τ

that harms cross-modal alignment. Furthermore, the results of *Prompt-Concise* show that further restricting the verbosity yields comparable results, indicating that our default 20-word constraint already provides an optimal balance between descriptive detail and noise control. These findings demonstrate that our prompt design effectively minimizes textual noise while preserving essential temporal cues.

Evaluation of Hyperparameters in Temporal Modeling. To analyze the effectiveness of the global and local branches in the query-guided temporal modeling module, conduct experiments using different weight parameter settings in Eq.(6), i.e., different values of hyperparameters a and b. The results are shown in Fig. 4. We observe that the combination of both global and local branches with equal weights ($a = b = 1$) yields the best performance. When the weights are unbalanced, the performance drops slightly. In addition, when only one branch is used, the performance tends to degrade.

Evaluation of Boundary Probability Modeling. To evaluate the effectiveness of the proposed boundary probability modeling strategy, we design four variants for comparison: (1) "Gauss distribution", which directly uses Gauss distribution to smooth the boundary annotation; (2) "Only distance", which only considers the temporal distance when calculating frame scores; (3) "Only similarity", which only considers the semantic similarity when calculating frame scores; (4) "Original query", which uses the original query to calculate similarity with video features when calculating frame scores. Table 5 shows the proposed strategy achieves the best results across multiple datasets, which suggests the effectiveness of introducing both the temporal distance and semantic similarity to the boundary probability modeling. Notably, since the query expansion is performed as an offline preprocessing step and cached for training, our module provides performance gains with negligible impact on training efficiency and additional inference latency.

Evaluation of Hyperparameters in Boundary Probability Modeling. We investigate the sensitivity of the model to the balancing coefficient α in Eq.(7) and Eq.(9), which controls the relative importance of temporal distance versus semantic similarity during pseudo boundary generation. As shown in Fig. 5, a smaller α (e.g., 0.5) relies more heavily on semantic alignment, which may lead to the selection of frames that are semantically similar but temporally distant from the ground truth, potentially introducing outliers. Conversely, a larger α (e.g., 1.5) imposes a stricter temporal constraint, which might limit the model's ability to adaptively find the true action boundaries when human annotations are imprecise. Experimental results indicate that $\alpha = 1.0$ provides the most robust performance, effectively balancing the global temporal prior with local semantic cues.

We also conduct experiments to analyze the impact of different τ values on model performance, as shown in Fig. 6. A lower τ (e.g., 0.5) results in more frames being included, which may introduce more noise. In contrast, a higher τ (e.g., 1.0) filters out low-confidence frames, reducing noise but potentially missing relevant boundary frames. The results suggest that an intermediate value $\tau = 0.8$ achieves the optimal balance, offering the best performance.

Evaluation of Robustness to Boundary Uncertainty. To evaluate the robustness of our method to boundary uncertainty, we introduce perturbations to the boundary anno-

Table 5 Results of different boundary probability modeling strategies with QD-DETR on the Charades-STA and TACoS datasets

Probability modeling	Charades-STA			TACoS		
	R1@0.5	R1@0.7	mAP	R1@0.5	R1@0.7	mAP
Gauss distribution	59.19	37.98	35.04	39.07	22.84	21.69
Only distance	59.11	37.34	35.44	40.19	23.09	21.70
Only similarity	59.35	38.04	35.33	39.64	22.94	21.43
Original query	59.35	38.33	35.61	39.67	23.22	21.87
Ours	**59.71**	**38.58**	**35.97**	**40.83**	**24.04**	**22.49**

The best results are highlighted in bold

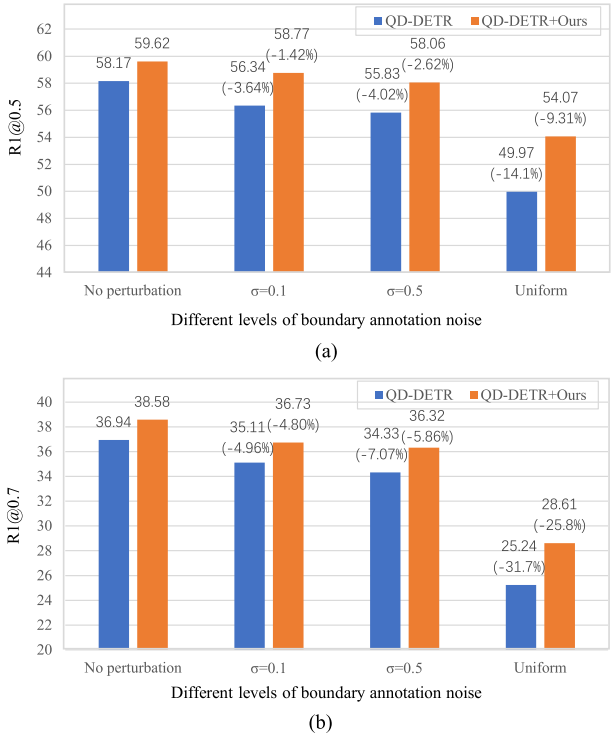


Fig. 7 Results of boundary annotation perturbation on the Charades-STA dataset by integrating our modules into QD-DETR

tations in the Charades-STA training set to increase the boundary uncertainty during training. The original boundary annotations τ_s, τ_e are perturbed to generate new annotations $\hat{\tau}_s, \hat{\tau}_e$, denoted by $\hat{\tau}_s = \tau_s + (\tau_e - \tau_s) \cdot X, \hat{\tau}_e = \tau_e + (\tau_e - \tau_s) \cdot Y$, where X and Y represent the distributions. We use three distributions for comparison: (1) "$\sigma = 0.1$", where $X, Y \sim N(0, 0.1^2)$; (2) "$\sigma = 0.5$", where $X, Y \sim N(0, 0.5^2)$; (3) "Uniform", where $X, Y \sim U(-0.5, 0.5)$. Note that these three distributions represent gradually increasing boundary annotation noise, which means that the uncertainty of boundary annotation increases. Fig. 7 shows that compared to QD-DETR, our method exhibits a relatively smaller performance drop as the level of annotation noise increases. Specifically, under a moderate perturbation of $\sigma = 0.5$, the R1@0.5 of base QD-DETR drops by 4.02%, while our framework only shows a 2.62% decrease. This gap becomes even more pronounced under the most severe *Uniform* perturbation, where the base model experiences a significant 14.10% drop in R1@0.5, whereas our method restricts this decay to 9.31%. A similar trend is observed in R1@0.7, where our model reduces the maximum drop from 31.67% to 25.84%. This demonstrates the robustness of our modules to boundary uncertainty, as they maintain a more stable performance profile even when the grounding supervision is highly noisy.

Evaluation of Robustness to Action Duration and Location. To further investigate the robustness of our framework across varying action durations and temporal positions, we conduct detailed subgroup analyses. For the action duration analysis, as shown in Table 6, we utilize the TACoS dataset due to its wide temporal span, categorizing samples in the test set into *Short* (the shortest 25%, < 6.8s), *Average*, and *Long* (the longest 25%, > 33.9s) subsets. Our framework achieves consistent performance gains across all duration scales. For the boundary-adjacent analysis, we evaluate our model on Charades-STA to leverage its high density of extremity-aligned samples. We define *Beginning* samples as actions starting at 0 s and *End* samples as those ending within 1 s of the video conclusion. As shown in Table 7, our method yields consistent improvements in both the *Beginning* (+0.31% R1@0.5) and *End* (+1.05% R1@0.5) scenarios. This confirms its ability to refine localization even when actions align with video limits.

4.5 User Study on Quality of Generated Queries

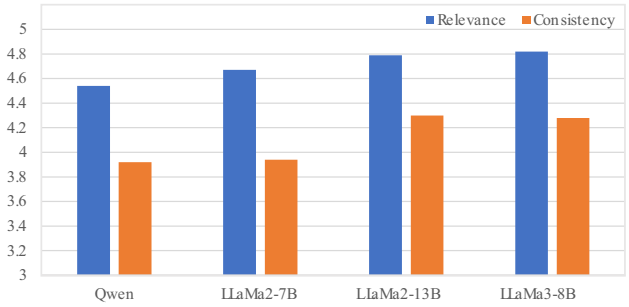
To evaluate the quality and consistency of action queries generated by different LLMs, we conduct a user study using 100 sets of queries generated by each model (Qwen, LLaMa2-7B/13B, and LLaMa3-8B) on the Charades-STA dataset. Each set consists of the original query and its corresponding expanded boundary queries (start and end). We assess each set using both human and automated evaluations along two dimensions: (1) relevance to the original query in the dataset, and (2) consistency between the two generated queries. To ensure fairness and minimize bias, we anonymize the source model of each query set and randomize their presentation order. For the human evaluation, five participants with substantial experience in video-language research (but not

Table 6 Performance comparison on different duration subsets on TACoS

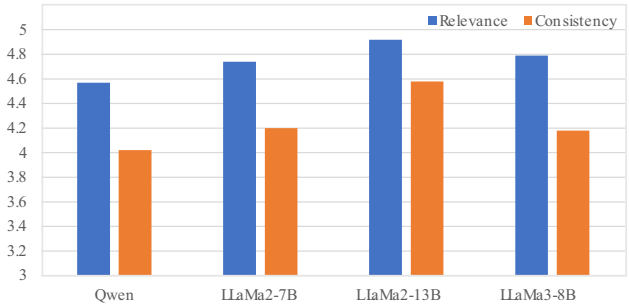
Subset	Size	Method	R1@0.5	R1@0.7	mAP
Short (< 6.8s)	998	QD-DETR	37.68	18.64	17.37
		+Ours	39.88	19.54	18.46
Average	2004	QD-DETR	44.61	25.10	23.31
		+Ours	46.86	28.99	26.37
Long (> 33.9s)	999	QD-DETR	26.13	12.41	16.62
		+Ours	30.13	14.51	18.30

Table 7 Performance comparison on different action location subsets on Charades-STA

Subset	Size	Method	R1@0.5	R1@0.7	mAP
Beginning	957	QD-DETR	70.85	48.90	44.42
		+Ours	71.16	51.93	45.25
End	665	QD-DETR	62.41	46.47	42.09
		+Ours	63.46	47.37	42.49



(a) GPT-based Evaluation Results



(b) Human Evaluation Results

Fig. 8 Quantitative evaluation of query expansion quality by GPT-4 and human raters

involved in our project) rate each presented query set alongside the corresponding video frames, and the final score is calculated as the average of their ratings. For the automated evaluation, we prompt GPT-4 to score each query set on a scale of 0 (very poor) to 5 (excellent) for both relevance and consistency.

As shown in Fig. 8a, GPT-4-based evaluation indicates that all models get relatively high scores, with larger models generally achieving higher relevance and consistency scores. The human evaluation results in Fig. 8b reveal a similar pattern, indicating the agreement between automated and human assessments. These results demonstrate the overall high quality and consistency of the queries generated by our method.

4.6 Qualitative Results

4.6.1 Consistent Localization Across Similar Queries

To qualitatively evaluate the effectiveness of our modules, we presents several examples of action localization results by integrating our modules into QD-DETR on the Charades-STA dataset. As shown in Fig. 9, we observe that for similar queries of "*close the door*", the start boundaries of the ground truth differ across examples: in Fig. 9 a and b, the annotations are at "*starting to pull the door*", while in Fig. 9c, the annotation is at "*reaching out to touch the door*". In contrast, we generate consistent expanded starting queries describing "*reaching out to touch the door*" for three examples. Consequently, our method predicts start boundaries consistently across examples, localizing them at "*reaching out to touch the door*", which also outperforms the predictions of QD-DETR.

4.6.2 Failure Case

In Fig. 9d, with the query of "person starts laughing", the action involves only subtle facial or body movements, making the transition into the action visually ambiguous. As a result, our model struggles to accurately determine the precise start point, highlighting the challenge posed by inherently ambiguous action boundaries.

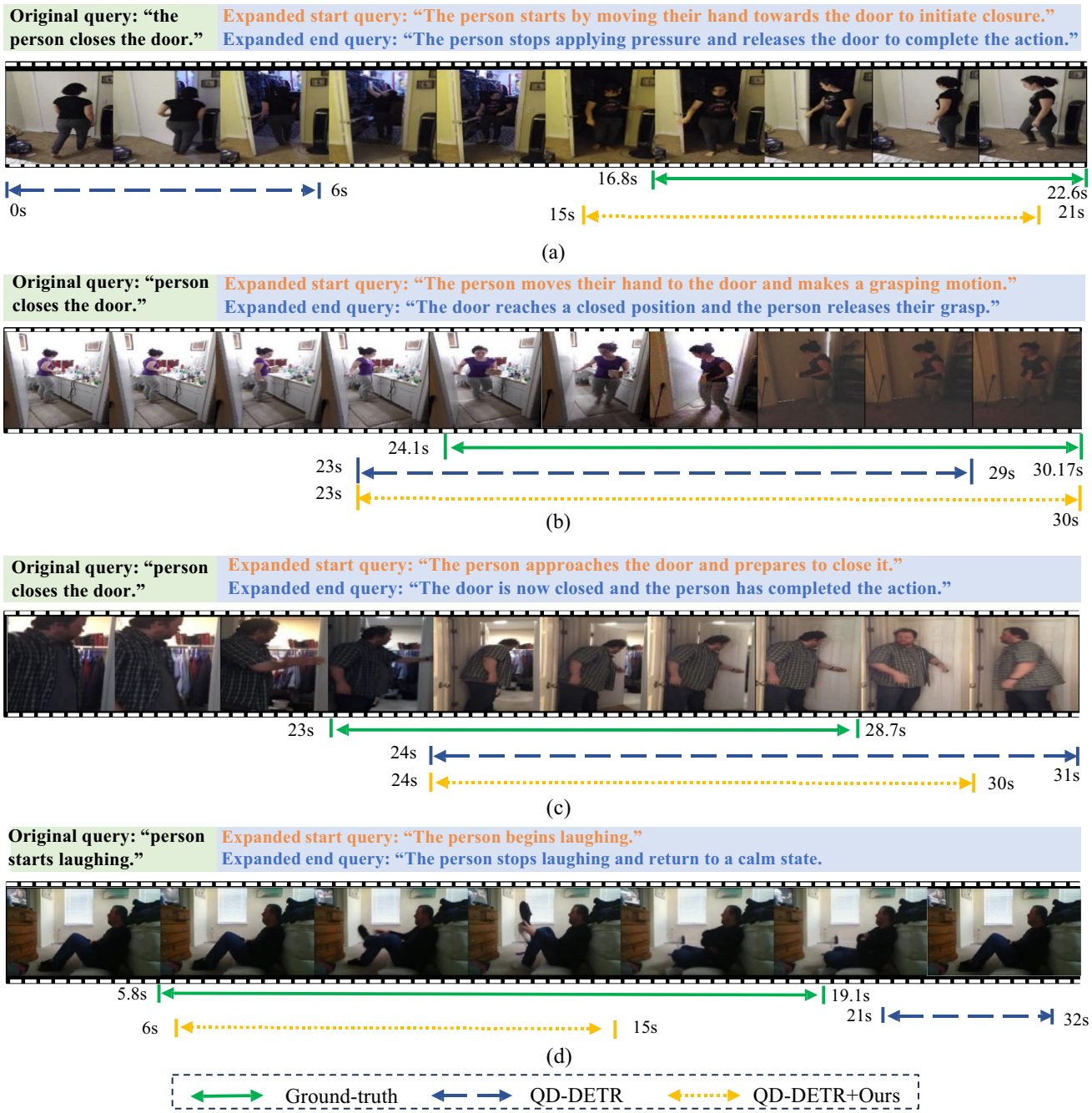


Fig. 9 Examples of action localization results with similar language queries on the Charades-STA dataset by integrating our modules into QD-DETR. The texts in red are the expanded start queries, and the texts in blue are the expanded end queries

4.6.3 Visualization of Boundary Probability Scores

To intuitively understand how our method models boundary uncertainty, we visualize the predicted boundary probability scores during training. As shown in Fig. 10, our method produces meaningful probability peaks at frames corresponding to the action boundaries described in the expanded queries, such as "*lifting their legs*". This indicates that the modeled probability scores effectively highlight the start and end of the action as guided by the expanded queries, providing robust boundary annotations.

4.7 Efficiency Analysis

To provide a comprehensive view for cold-start scenarios where real-time expansion is required, we report the end-to-end cost on TACoS in Table 8. The results show that while the LLM expansion (LLaMa3-8B) introduces an initial computa-



Fig. 10 Visualization of boundary probability scores on the Charades-STA dataset by integrating our modules into QD-DETR

Table 8 Inference latency and computational cost (per sample) on the TACoS dataset

Process	Time (ms)	FLOPs (G)
Query Expansion	~1315	1442.7
Visual Encoder	~1344	54.9
Text Encoder	~2.90	1.38
QTM	~19.83	3.09
Localization Head	~17.87	2.46

tion, its impact on the overall end-to-end latency is negligible. Crucially, the Query Expansion and Visual Encoding processes are executed in parallel. For a video with the average duration of the TACoS dataset (~4.7 min), the visual encoding (SlowFast+CLIP) takes approximately 1.34s on an RTX 4090 (averaging 9.53 ms per 2 s clip), which is comparable to the LLM's expansion time (~1.31s). Thus, the expansion process is effectively 'hidden' within the video processing pipeline, incurring zero additional time cost to the total system latency. In this context, the one-time semantic enrich-

ment provided by the LLM is a worthwhile trade-off for the substantial localization gains.

5 Conclusion

We have presented a large language model (LLM)-powered query expansion method to enhance boundary prediction for language-driven action localization. By leveraging LLMs to generate expanded boundary queries that provide cues for start and end boundaries, our method captures boundary information that original language queries overlook. Additionally, the boundary probability modeling transforms rigid boundary annotations into probability scores, leveraging the semantic similarities between frames and the expanded query, as well as the temporal distances between frames and the annotated boundary frames, to provide more flexible and consistent supervision. The proposed modules are model-agnostic and can be seamlessly integrated into existing models of language-driven action localization in an off-the-shelf manner. Extensive experiments on five state-of-the-art models across three datasets validate the effectiveness of our method in reducing the impact of boundary uncertainty and enhancing boundary prediction.

Author Contributions Shuo Yang is the corresponding author. The individual contributions are as follows: Conceptualization: Zirui Shang, Xinxiao Wu, and Shuo Yang. Methodology and Software: Zirui Shang. Formal Analysis and Investigation: Zirui Shang. Writing—Original Draft: Zirui Shang. Writing—Review & Editing: Xinxiao Wu and Shuo Yang. Supervision: Xinxiao Wu and Shuo Yang. Funding Acquisition: Xinxiao Wu and Shuo Yang.

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Data Availability All datasets used in this study are open access and have been cited in the paper, i.e. TACoS (https://www.coli.uni-saarland.de/projects/smile/page.php?id=tacos), QVHighlights (https://github.com/jayleicn/moment_detr) and Charades-STA (https://prior.allenai.org/projects/charades).

Declarations

Conflict of Interest The authors have no conflict of interest to declare that are relevant to the content of this article.

Ethical Approval Not applicable.

Consent to Participate Informed consent was obtained from all individual participants involved in the user study. All participants were fully informed of the study's purpose and participated voluntarily.

Consent for Publication Not applicable.

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